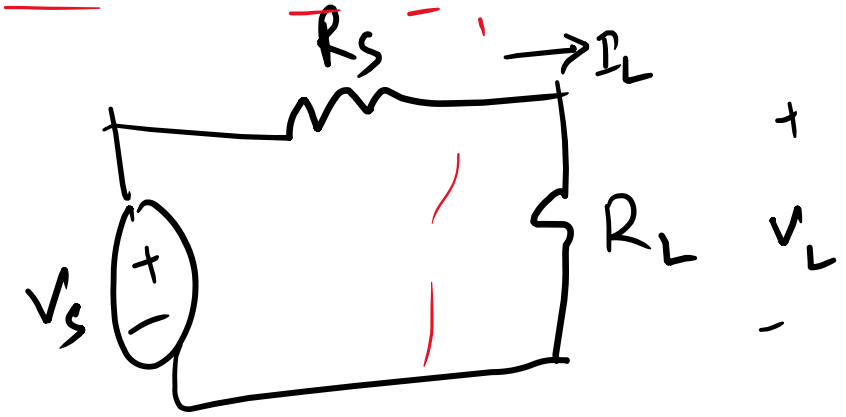
Maximum Power Transfer theorem



$$P_L = I_L^2 \cdot R_L = \frac{V_S^2}{(R_S+R_L)^2} \cdot R_L$$

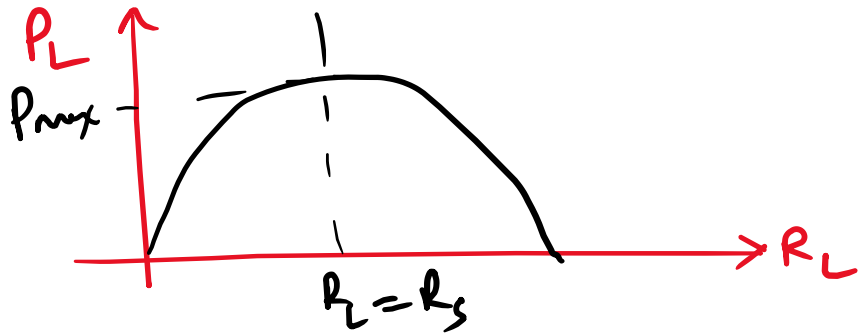
$$I_L = \frac{V_S}{R_S+R_L}$$

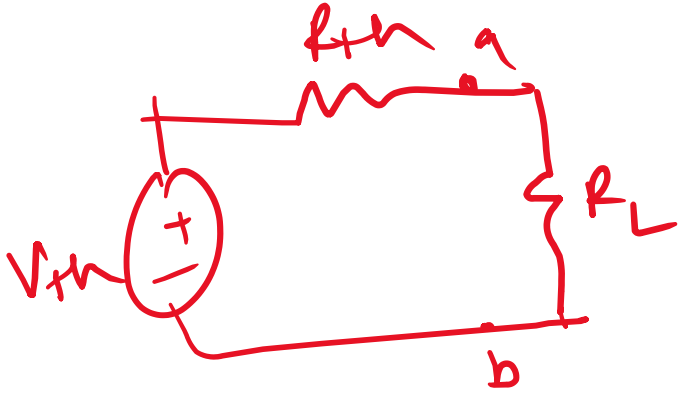
Find R_L so that max power is transferred from source to load.

$$\frac{dP_L}{dR_L} = 0 = \frac{(R_S+R_L)^2 \cdot V_S^2 - V_S^2 \cdot R_L(2)(R_S+R_L)}{(R_S+R_L)^4}$$

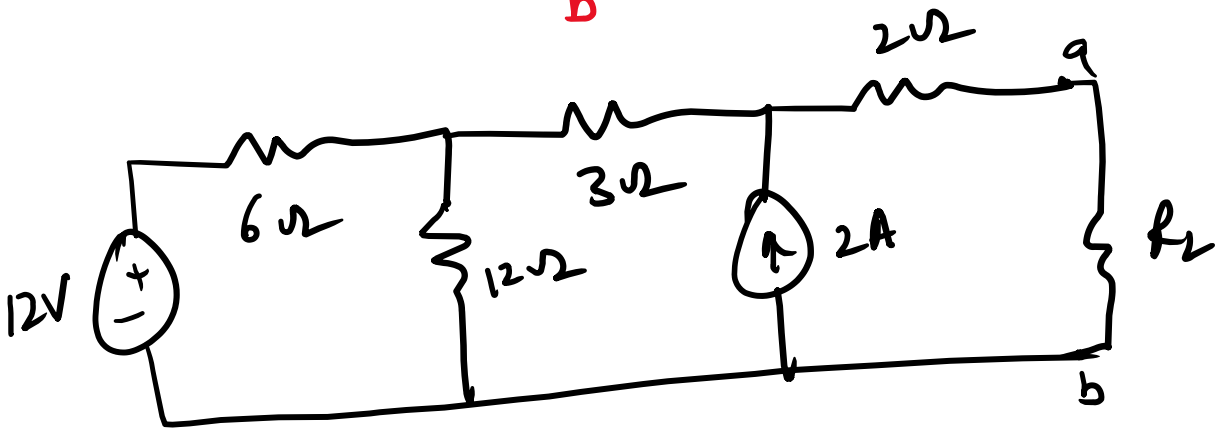
$$2 \cdot R_L (R_S+R_L) = (R_S+R_L)^2 \rightarrow R_L = R_S$$

Max power is transferred from source to load when load resistance is equal to source resistance.





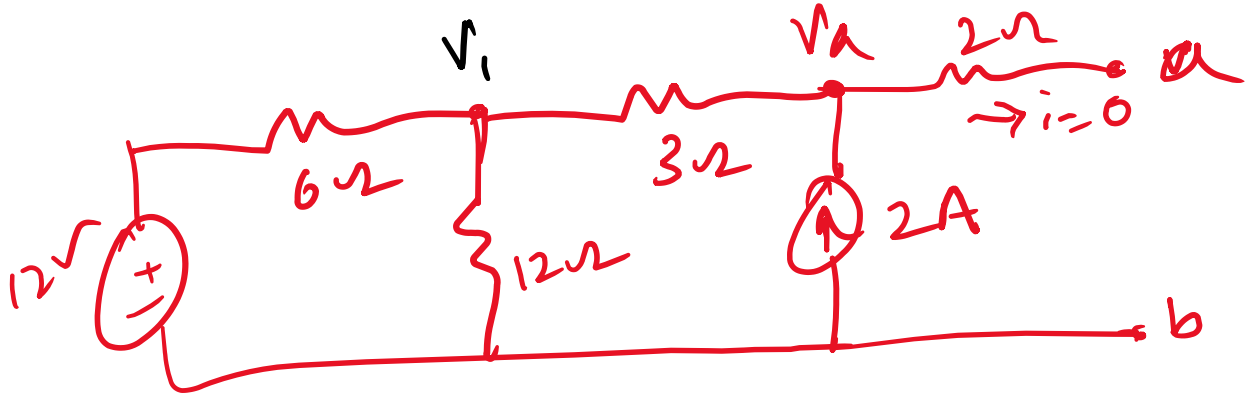
Max power when $R_L = R_{th}$.



Find value of R_L for max power transfer.

Also find the value of max power.

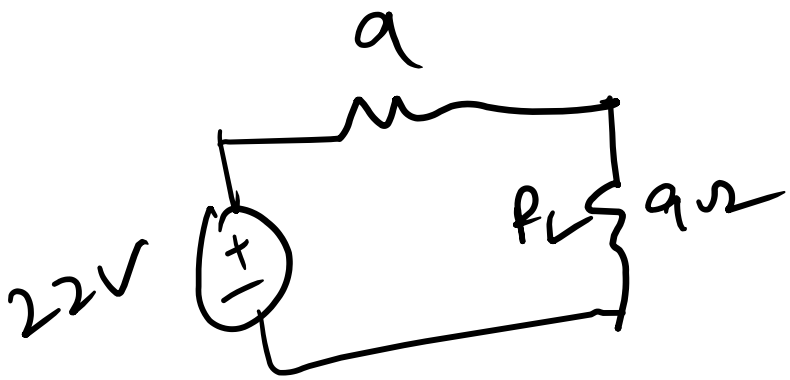
$R_{th} =$ 9Ω



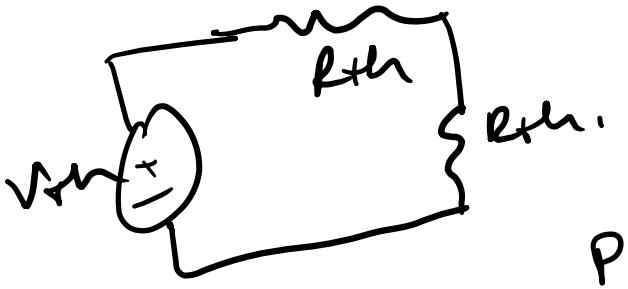
$$\frac{12-V_1}{6} = \frac{V_1}{12} + \frac{V_1-V_a}{3}$$

$$\frac{V_1-V_a}{3} + 2 = 0$$

$$V_a = V_{th} = 22V$$



$$P = \frac{22^2}{4 \cdot 9} = 13.44W$$



$$i = \frac{V_{th}}{2R_{th}}$$

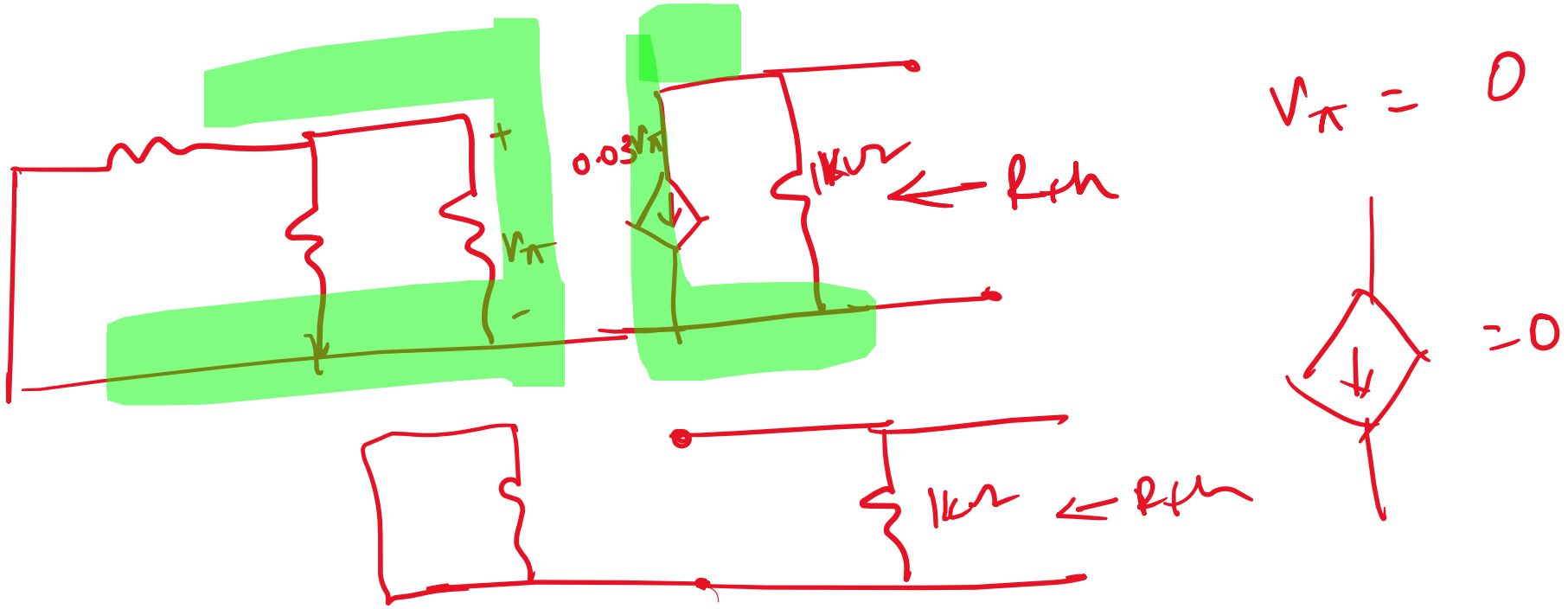
$$P = \frac{V_{th}^2}{4R_{th}^2} \cdot R_{th}$$

$$P = \frac{V_{th}^2}{4 \cdot R_{th}}$$



C-E B.J.T. model.

Choose a load resistance so m.p.t. from amplifier to load



$R_{th} = 12\Omega$